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WP4 – Geometric / Cohomological Lifts in Triatomic and Three-Body Systems

Triatomic Holonomy Reduction and Principal S^1 Bundles

Holonomy and S^1-Bundle Structure: A triatomic (three-body) molecular system provides a canonical example of an S^1-bundle arising from symmetry reduction. After eliminating the center-of-mass motion, the *translation-reduced* configuration space P for three atoms is six-dimensional ($\mathbb{R}^3 \times \mathbb{R}^3$ minus collinear points) on which $SO(3)$ acts freely by overall rotations [1]. This $SO(3)$ principal bundle $P\to M$ has base **shape space** $M \cong \mathbb{R}^3_{+}$ (the space of internal Jacobi coordinates, e.g. two bond lengths and the included angle) [1]. The kinetic energy metric induces a natural **mechanical connection** on P (horizontal subspaces orthogonal to rotation orbits). Importantly, this connection has a **nontrivial holonomy**: parallel transport of the molecule's orientation around certain loops in shape space results in a net rotation (an "anholonomy"). In fact, the holonomy group for triatomic molecules is $SO(2)$ [2], meaning there is effectively a one-parameter residual rotation (about a fixed axis) that cannot be eliminated globally. Using the holonomy reduction theorem, one **reduces P further to an S^1 principal bundle** Q (the *holonomy bundle*), which is topologically $M \times S^1$ [3]. In other words, we partially fix the orientation (up to a remaining S^1 freedom) such that $Q \cong \mathbb{R}^3_{+} \times S^1$ is a trivial bundle [4]. The S^1 fiber corresponds to the *internal rotation angle* (or *gauge angle*) associated with the holonomy.

Extended Phase Space and Conserved Momentum: Passing to phase space, the reduced bundle Q yields an extended phase space T^*Q for the internal dynamics. This extension includes the extra cyclic coordinate (the S^1 angle, often denoted γ) and its conjugate momentum p_γ. Crucially, p_γ is conserved (it generates the S^1 symmetry) and serves as an S^1-equivariant momentum map [5]. Physically, p_γ represents the angular momentum about the remaining gauge axis (or another conserved quantity analogous to a "charge"). The Hamiltonian on T^*Q can be derived explicitly. For example, in planar Jacobi coordinates one finds a kinetic energy of the form (for some internal coordinates r,s,ϕ and the gauge angle γ):

$$H = \tfrac{1}{2}(p_r^2 + p_s^2) \;+\; \tfrac{1}{2}\Big(\frac{1}{r^2} + \frac{1}{s^2}\Big)p_\phi^2 \;-\; \frac{1}{r^2}\,p_\phi\,p_\gamma \;+\; \frac{1}{2r^2}p_\gamma^2 \;+\; V(r,s,\phi)\,.$$

This Hamiltonian includes a **cross-coupling term** $-\,\frac{1}{r^2}p_\phi\,p_\gamma$ between the vibrational angular momentum p_ϕ and the gauge momentum p_γ [6]. Such a term is the hallmark of a *magnetic coupling* induced by the curvature of the connection. Indeed, if p_γ is treated as a constant (since γ is cyclic), the reduced dynamics in the (r,s,ϕ) subspace feels an effective **vector potential** $\sim p_\gamma/r^2$ – analogous to a charged particle moving in a magnetic field.

Mechanical Connection and Curvature: Geometrically, the mechanical connection on P yields a connection 1-form A on Q whose curvature $F=dA$ manifests as this effective magnetic field on shape

space. The S^1 fiber coordinate γ plays the role of a gauge angle with A as its gauge potential. The non-zero curvature $F\neq0$ is responsible for the **anholonomy** (geometric phase) in internal motions. After performing symplectic reduction by the S^1 (i.e. fixing p_γ), the remaining dynamics on shape space is governed by a *twisted* symplectic form containing the curvature term. In the triatomic case, one finds that the **holonomy-reduced** configuration space Q (with an S^1) makes the system equivalent to a particle moving in $\mathbb{R}^3$ under a magnetic field. In fact, the holonomy-reduced space $\mathbb{R}^3_{+}\times S^1$ and its conserved momentum allow an exact identification: **the triatomic molecule's reduced motion is equivalent to a charged particle on $\mathbb{R}^3$ with a fixed "charge" p_γ producing a magnetic field** [7] [8]. This is a classical analog of a Kaluza–Klein construction: by extending $\mathbb{R}^3$ to $\mathbb{R}^3 \times S^1$ and treating p_γ as a charge, the complicated internal dynamics (with Coriolis-like forces) is lifted to free or simpler dynamics in a higher-dimensional space [7].

Key references: The above picture is drawn from the work of Iwai, Littlejohn & Reinsch, and others on molecular dynamics. In particular, Littlejohn and Reinsch (1997) laid out the gauge theory of molecular rotation–vibration coupling, showing how a mechanical connection on $SO(3)$-bundles leads to geometric phases. Recent explicit reductions for triatomic systems confirm that $Q = \text{Shape}\times S^1$ and detail the reduced Hamiltonian with its magnetic term [2] [6]. This S^1 holonomy bundle construction is special to three-body systems – for four or more bodies, the holonomy group becomes $SO(3)$ (not just S^1) and the bundle is non-trivial [9], complicating a similar reduction.

Symplectic Regularization and S^1 Bundles in the Three-Body Problem

Parallel to the above *geometric holonomy* approach, classical celestial mechanics has long utilized **regularization** methods that introduce extra S^1 symmetries. These methods, pioneered by Levi-Civita, Kustaanheimo-Stiefel (KS), and Moser, aim to *lift* the problem to a higher-dimensional phase space where singularities are resolved and hidden symmetries emerge.

- **Kepler (Two-Body) Regularization:** The Kepler problem (which is the sub-problem of two-body motion within the three-body system) famously admits an S^1 (actually $SO(2)$ or double-cover S^1) symmetry upon regularization. Moser's regularization shows that a negative-energy Kepler motion on the plane can be transformed such that the **energy hypersurface compactifies to $\mathbb{R}P^3$**, which is topologically a circle bundle over S^2 (the Hopf fibration) [10]. For the spatial Kepler problem, the energy surface is S^3 (an S^1 bundle over S^2) due to the introduction of the KS 4-dimensional spinor and an associated $SU(2)$ (double-cover of $SO(3)$) symmetry. In each case, the act of regularization introduces an extra phase angle (half of the true anomaly, or a frame rotation angle) that plays the role of a **bundle coordinate** (often interpreted as *fictitious time* or a gauge angle). The physical trajectory is obtained by projecting from the lifted flow in the S^1-extended phase space down to the original space. The additional S^1 symmetry yields a conserved quantity (conjugate momentum), and reducing by this symmetry recovers the original dynamics. Notably, the symplectic form in regularized coordinates picks up a **curvature term** corresponding to the so-called **"geodesic" flow on S^2 with a magnetic monopole** – a viewpoint articulated by Montgomery and others for the Kepler problem.

- **Restricted Three-Body Problem (CR3BP):** Modern symplectic treatments of the circular restricted 3BP likewise reveal hidden S^1 bundle structures. For example, when considering collisions of the small mass with one of the primaries, one can apply a Moser-type mapping in local coordinates. The result is that each collisional energy manifold, after regularization, becomes a product involving S^3 or $\mathbb{R}P^3$. In fact, for the spatial CR3BP, the energy hypersurface near a single collision can be compactified to **$S^3 \times S^2$** (and in more complex scenarios, connected sums of such products) [10]. Here S^3 (or RP^3 in the planar case) carries the structure of a principal S^1 bundle over S^2 – essentially the same Hopf fibration arising from the two-body subproblem. Each S^3 factor corresponds to an **SU(2) / S^1 symmetry** introduced via regularization of a two-body collision. Thus, the restricted 3BP with both primaries introduces two such bundle structures (one for each primary-small body collision), reflected in the product $S^3 \times S^2$. These bundle symmetries are typically **global** (they exist on the compactified energy manifold) but may not commute with the full dynamics unless energy is fixed.

- **Elimination of the Nodes (Partial Reduction):** There is also a classical S^1 symmetry that appears when reducing the rotational symmetries of the three-body problem *without* fully regularizing collisions. In the planar problem, the total angular momentum about the center of mass is conserved; fixing this and eliminating the corresponding rotational coordinate (Jacobi's elimination of the node) effectively removes an S^1 symmetry. However, as noted by Smale (1983) and others, after this elimination the **symplectic structure is twisted by a magnetic term** [11]. That is, the act of removing the node (an S^1 action) introduces additional terms in the reduced 2-form, analogous to the gauge field effects seen in the triatomic case. In contemporary geometric language, the reduced phase space (with one rotation angle removed) carries a *modified* symplectic form $\omega_{\text{red}} = \omega_{\text{naive}} - \beta$, where β is an added **curvature two-form**. This β is precisely the curvature of the mechanical connection associated with the eliminated S^1 (rotation about the reference axis). In general, whenever a continuous symmetry is factored out of a Hamiltonian system, if the symmetry's momentum map μ is nonzero (or if a gauge choice breaks global triviality), one obtains a **"magnetic" 2-form** on the reduced space given by the curvature of the connection used for reduction [12]. In the three-body context, this phenomenon appears in both astronomy (Jacobi coordinates with node elimination) and molecular dynamics (body-frame elimination in polyatomic molecules). It highlights that a principal bundle viewpoint (with base = shape space, fiber = symmetry group) is natural for understanding the reduced three-body dynamics.

Summary of S^1 Bundle Occurrences: Across these examples, the three-body problem (and its sub-cases) exhibits S^1 bundles in two main ways: (1) **Regularization lifts** (Levi-Civita or KS type) introduce an *auxiliary S^1 phase* to desingularize collisions, leading to Hopf-fibration structures (e.g. $S^3 \to S^2$) and hidden symmetries; (2) **Rotational reductions** (fixing angular momentum or frames) naturally produce an S^1 fiber (the remaining trivial rotation or gauge angle), whose connection curvature yields extra terms in the reduced dynamics ("Coriolis forces" or geometric phases). Both are mathematically analogous: in each case an extended phase space with an S^1 symmetry is constructed so that the physical system is a **symplectic reduction** of that extended system.

Mechanical Connections and Magnetic Curvature in Reduced Dynamics

A unifying theme in these studies is the role of the **mechanical connection** and its curvature form as the source of "magnetic" forces in reduced systems. In a principal G-bundle setting (with G compact), the mechanical connection 1-form $\mathcal{A}$ on the configuration bundle encapsulates how the system's symmetry coordinates (e.g. Euler angles, gauge phases) relate to the shape coordinates. When one performs symplectic reduction by G, any *non-zero curvature* $F = d\mathcal{A}$ appears in the reduced phase space as an additional closed 2-form (often adding to the canonical symplectic form). If $G = S^1$ (an Abelian symmetry), this curvature 2-form is an exact *magnetic field* (locally $d\mathcal{A}$) on the shape space. In physical terms, the reduced system behaves like a charged particle moving in a magnetic field defined by F. We have already seen this in the triatomic molecule: the curvature of the $SO(3)\to S^1$ connection produces the $-p_\phi p_\gamma/r^2$ coupling in the Hamiltonian and an induced Lorentz-force term in the equations of motion [6]. From a Hamiltonian perspective, one can say that the **reduced symplectic form** is twisted by the pullback of this curvature. Marsden and coworkers explicitly note that in such reductions *"the magnetic term is the curvature of the mechanical connection"* [12]. Thus, understanding the connection on the S^1-bundle is key to understanding the non-integrability or extra precession in the reduced three-body dynamics.

Concretely, if θ is the S^1 fiber coordinate and p_θ its conjugate momentum (momentum map), the symplectic form on the extended space $T^*(\text{Shape}\times S^1)$ can be written (in a local trivialization) as $\widetilde{\omega} = \omega_{\text{shape}} + d\theta \wedge dp_\theta$. Imposing a momentum level $p_\theta = \mu$ and reducing, one obtains $\omega_{\text{red}} = \omega_{\text{shape}} - \mu\, d\mathcal{A}$ on the shape space (since $d\theta$ is replaced by the connection form $\mathcal{A}$ upon identifying horizontal lifts). The term $-\mu\,d\mathcal{A}$ is the magnetic curvature term. If $d\mathcal{A}$ is exact (as in simple Abelian cases, where one can write $d\mathcal{A}=d(d\chi)$ for some angle function χ), this is a trivial shift; but in general (especially for non-Abelian or topologically nontrivial bundles) the curvature represents a real geometric obstruction. In celestial mechanics language, these curvature terms correspond to Coriolis and centrifugal effects* that cannot be removed by any global canonical transformation once the symmetry is fixed. They are precisely what one must "lift" if seeking an integrable extension of the system.

Toward an S^1-Lift for the Three-Body Problem

The insights from triatomic molecules and regularized two-body systems strongly support the conjecture that a **minimal S^1 lift** could clarify the three-body problem's dynamics. The idea is to find an extended phase space $\widetilde{M}$ that includes one additional cyclic coordinate (or perhaps a few) such that the complicated reduced dynamics becomes a simple flow (possibly integrable or at least easier) upstairs. Based on the literature:

- In the triatomic case, $\widetilde{M}$ was effectively $T^*(\mathbb{R}^3_{+} \times S^1)$, and indeed the dynamics on T^*Q was a **natural mechanical system** (cotangent bundle with standard form) [13]. The physical effects of curvature only appeared after projecting back down. This matches the **S^1 minimal lift strategy**: introduce an S^1 so that upstairs the symplectic form is untwisted and the Hamiltonian has no magnetic terms, while downstairs the S^1's curvature accounts for the missing forces.

- By analogy, for the planar three-body problem (or restricted 3BP), one can hypothesize $\widetilde{M} = T^*(\mathcal{S} \times S^1)$ where $\mathcal{S}$ is the shape space of the three-body configuration (e.g. the space of triangles, modulo similarity if scaling out total moment of inertia). Here the extra S^1 coordinate would relate to a symmetry in the extended system. A natural guess is that this S^1 corresponds to the Jacobi node angle* or an equivalent phase that, when not fixed, restores a hidden symmetry. For instance, in the full 3D problem, one might require an $SO(3)$ or $SU(2)$ extension (as KS suggests); but in a simpler planar or spatially-symmetric context, a single S^1 might suffice (e.g. representing the rotation of the triangle's reference frame about an axis). The momentum conjugate to this S^1 would then be a constant (perhaps related to total angular momentum or another integral), and the zero-level of its momentum map would impose a constraint akin to the classical "angular momentum = constant" or KS bilinear constraint.

- The **mechanical connection** on the existing rotation bundle in the three-body problem provides a concrete candidate for this S^1. In the planar case, after eliminating the center-of-mass, the configuration space is $\mathbb{R}^4$ (two relative position vectors); eliminating the overall rotation $SO(2)$ leaves an S^1 worth of gauge freedom (the node), exactly as in the elimination-of-nodes procedure. *Re-introducing* that S^1 as a physical coordinate (instead of reducing it out) means considering an extended phase space where the node angle is retained as a dynamical coordinate and its conjugate momentum is constant (equal to the total angular momentum). This is precisely the extension that yields the **magnetic term** upon reduction. In a sense, the extended system that *keeps* the node angle (or an equivalent phase) will have a trivial symplectic structure (no magnetic term) but one extra degree of freedom and symmetry.

- One can draw a direct line from the triatomic holonomy example to a potential 3BP lift. In triatomics, $Q = \text{Shape}\times S^1$ and the final reduced space was $Q/S^1 = \text{Shape}$ with curvature-induced forces. For the Newtonian three-body problem, one may seek a similar $Q = \text{Shape}\times S^1$ such that Q/S^1 gives the usual shape dynamics (with its troublesome curvature). The challenge is to identify the correct connection and group action so that the **algebraic constraints** defining Q correspond to a **momentum map level set**. The KS transformation for two-body is a precedent: the quadratic constraint $|u|^2 = r$ defining the 4D spinor sphere S^3 is the zero-level set of an $SO(2)$ moment map (it imposes a fixed $U(1)$ charge in the extended oscillatorial system). By analogy, a three-body S^1 lift would have constraints (perhaps one per pair, or related to the triangle's moments of inertia) that can be seen as momentum map conditions for some torus action.

In summary, *WP4's literature* shows that **geometric lifts using S^1 (and higher) bundles are well-established** in three-body-like systems: triatomic molecules use an S^1 holonomy bundle to encode vibrational phases, and celestial mechanics uses S^1 (or S^3) bundles to regularize or partially fix the three-body problem. The curvature of the mechanical connection in these bundles introduces exactly the "magnetic" terms that distinguish the reduced (physical) dynamics from a simple integrable model. This strongly endorses our conjecture that *the right extended phase space for the 3BP will involve a principal S^1 (or possibly a product of circles) whose momentum map constraints project to the necessary KS-type algebraic conditions.* Finding this construction – essentially, a unification of the KS regularization with the mechanical connection framework – is the next step toward uncovering hidden structure in the three-body problem. The literature to date provides the pieces: principal bundle reductions, momentum maps, holonomy-induced forces, and analogies to charged-particle dynamics [7] [8]. The task now is to assemble these pieces into a concrete lift for the full three-body problem, verifying the conjecture that *the KS constraints are*

momentum map conditions on a higher-dimensional oscillator, with the physical three-body orbits emerging via symplectic reduction of that extended system.

References: The concepts in this WP are drawn from interdisciplinary sources in molecular dynamics and classical mechanics. Notable works include Guichardet's analysis of molecular rotation-induced gauge fields, Littlejohn & Reinsch's **Rev. Mod. Phys.** review on the gauge theory of molecular vibrations [2], Iwai's and Kazuo's papers on $R^3\times S^1$ Kaluza–Klein models for charged particles [7], Smale's and Marsden's expositions of node elimination and magnetic terms in N-body mechanics [12], and Agustin Moreno's recent symplectic treatment of the CR3BP [10]. These works collectively illuminate how adding a well-chosen extra phase (an S^1) can simplify dynamics at the expense of introducing a curvature that accounts for previously inexplicable behavior in the reduced (physical) system. The goal of WP4 has been to synthesize these insights as a stepping stone toward a full *geometric regularization* of the three-body problem.

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